



TEST REPORT Doc. EMC-VRP-ENERGIS-10IN-MPDU

Manufacturers Verification Report for Electro Magnetic Compatibility, according to The EMC Directive 2014/30/EU, Annex II, Module A, 2. Electromagnetic compatibility assessment in regards of radiated emission, excluding immunity tests. (EMC screening)

Report Information	Details
Report Title:	EMC Test Report according to EN IEC 61000-6-3
Product Under Test (PUT):	ENERGIS-10IN-MPDU
Applied Standard:	EN 61000-6-3:2007 + A1:2011 (IEC 61000-6-3:2006 + A1:2010)
Test Category:	Managed distributed power delivery unity
Report Revision/STATE:	1
Date of Issue:	29.03.2026

Organization Information	Details
Manufacturer:	Private person – David Sipos
Applicant:	David Sipos
Test Laboratory:	INDES EMC e.U. Zetschegasse 3, 1230 Wien, Austria
Test Date(s):	29.03.2026

Harmonized Standard	Final Verdict	Categorization	Reference
EN 61000-6-3	PASS	N/A	3.1

History of Changes

Date	Author	Revision	Modification
29.03.2026	Aleksandar Sindrak/ INDES EMC	1	Initial Version

Signature Page

Authorized Person(obligatory):

N/A_____ (Not applicable for pre- Verification report)

Name, Surname, Title

Test Conductor(obligatory):

N/A_____ (Not applicable for pre- Verification report)

Name, Surname, Title

Test Service Partner (optional):

N/A_____ (Not applicable for pre- Verification report)

Ing. Aleksandar Sindrak

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1. General Information

1.1 Equipment under test (EUT)

EUT:	ENERGIS – 10IN MANAGED PDU
Product Identification:	N/A
Product Revision:	Revision 1
Product Fitting Variant:	Representative functional prototype
Serial number/s:	0167665
Date of manufacturing:	Q4/2025

Manufacturer:	David Sipos
Address:	N/A

Description of the EUT: The 10IN MANAGED PDU is a power distribution device that delivers from an AC inlet power to 8 output sockets. It is intended to be used in light industrial/residual environments and is rack mounted.



Figure 1: ENERGIS 10IN MANAGED PDU

1.2 Test Setup Configurations

1.2.1 Conducted Emission Setup 150kHz – 30MHz

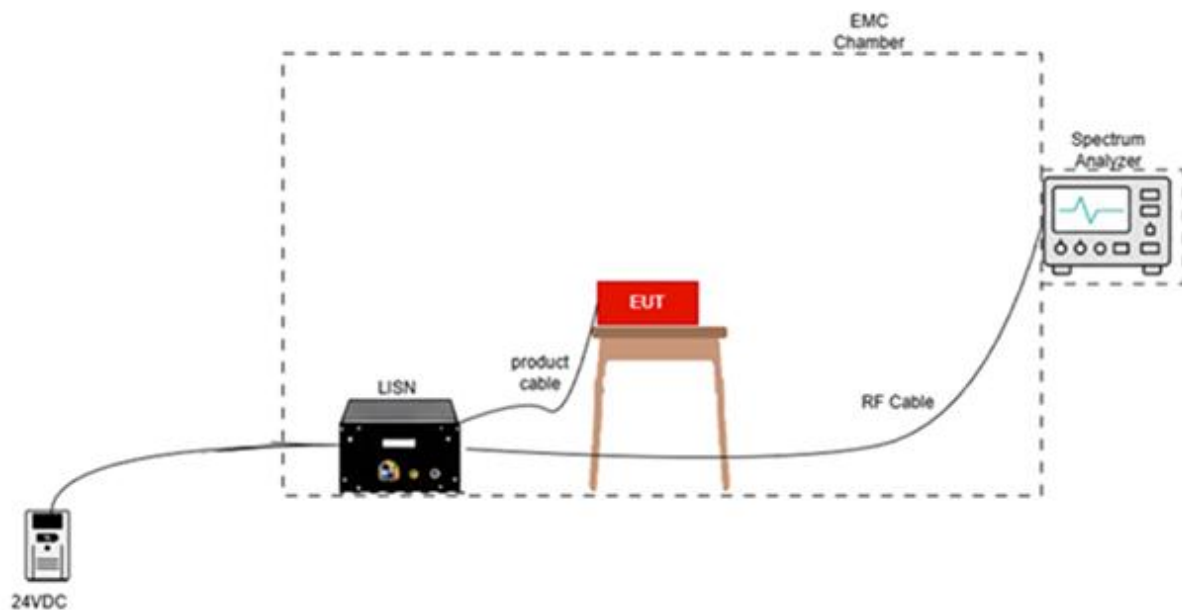


Figure 2: Conducted Emission Setup 150kHz – 30MHz

1.2.2 Radiated Emission Setup 30MHz – 1000MHz

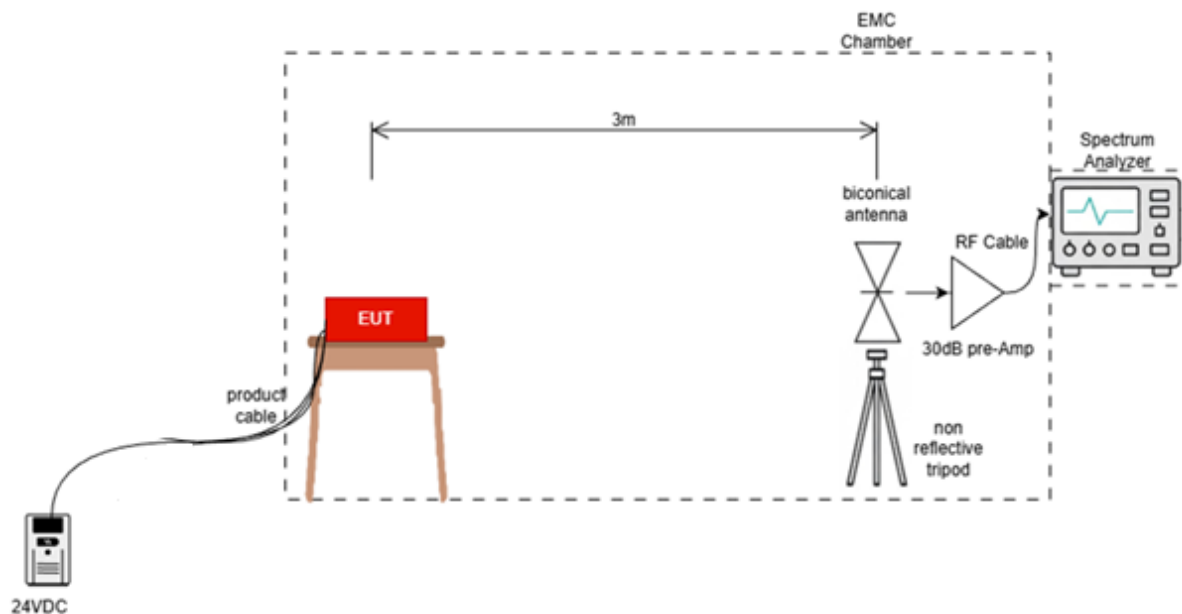


Figure 3: Radiated Emission setup 30MHz - 1000MHz

2. Test Plan

2.1 Applicable emission limits according to IEC 61000-6-3

2.1.1 Conducted emission Limits

2) Low voltage AC mains port	0 kHz – 2 kHz		IEC 61000-3-2 IEC 61000-3-3 IEC 61000-3-11 IEC 61000-3-12	See Notes 2 and 3.	
	0,15 MHz – 0,5 MHz	66 dB(μV) – 56 dB(μV) quasi-peak 56 dB(μV) – 46 dB(μV) average	CISPR 16-2-1, 7.4.1 CISPR 16-1-2, 4.3	See Notes 4 and 5.	
	0,5 MHz – 5 MHz	56 dB(μV) quasi-peak 46 dB(μV) average			
	5 MHz – 30 MHz	60 dB(μV) quasi-peak 50 dB(μV) average			
	0,15 MHz – 30 MHz	See basic standard, clause: discontinuous disturbance.	CISPR 14-1		

NOTE 1 If the internal emission source(s) is operating at a frequency below 9 kHz then measurements need only to be performed up to 230 MHz.

NOTE 2 Applicable to apparatus covered within the scope of IEC 61000-3-2, IEC 61000-3-3 or IEC 61000-3-12.

NOTE 3 If the apparatus is covered by the scope of IEC 61000-3-11, this may be used instead of IEC 61000-3-3.

NOTE 4 At transitional frequencies the lower limit applies.

NOTE 5 The limits decrease linearly with the logarithm of the frequency in the range 0,15 MHz to 0,5 MHz.

NOTE 6 The current and voltage disturbance limits are derived for use with an impedance stabilization network (ISN) which presents a common mode (asymmetric mode) impedance of 150 Ω to the telecommunication port under test (conversion factor is $20 \log_{10} 150 / 1 = 44$ dB).

Figure 4: Extract IEC 61000-6-3, Table 1

2.1.2 Radiated emission Limits

Table 1 – Emission

Port	Frequency range	Limits	Basic standard	Applicability note	Remarks
1) Enclosure port – Open area test site or semi-anechoic method	30 MHz – 230 MHz 230 MHz – 1 000 MHz	30 dB(μV/m) Quasi-peak at 10 m 37 dB(μV/m) Quasi-peak at 10 m	CISPR 16-2-3	See Notes 1 and 4.	May be measured at 3 m distance using the limits increased by 10 dB.

Figure 5: Extract IEC 61000-6-3, Table (continued)

2.2 Pre assessment Test plan for the EUT in accordance of IEC 61000-6-3

Test Index	Definition	Limit/Severity	Performance Criterion
CE1	Conducted emission of an AC power port	0,15 MHz – 0,5 MHz 66 dB(μV) – 56 dB(μV) quasi-peak 56 dB(μV) – 46 dB(μV) average 0,5 MHz – 5 MHz 56 dB(μV) quasi-peak 46 dB(μV) average 5 MHz – 30 MHz 60 dB(μV) quasi-peak 50 dB(μV) average	N/A
RE1	Radiated emission of EUT acc. IEC 61000-6-3 at 3m distance to Antenna	30 – 230 MHz: 40dB(μV/m) QP 230 – 1000 MHz: 47dB(μV/m) QP	N/A

Table 1: Test Plan for the EUT

3. Test Results

3.1 Test Results overview

Test Index	Definition	Reference	Performance Criterion	Verdict
CE1	Conducted emission of an AC power port	3.2.1	N/A	PASSED
RE1	Radiated emission of EUT acc. IEC 61000-6-3 at 3m distance to Antenna	3.2.2	N/A	PASSED

Table 2: Test Result overview for the EUT

3.2 Detailed Test Results

3.2.1 CE1 – Conducted Emission

3.2.1.1 Test Setup

See 1.2.1 Conducted Emission Setup 150kHz – 30MHz for test setup arrangement.

3.2.1.2 Test Level

Test Method	Level
Conducted emission over LISN	0,15 MHz – 0,5 MHz 66 dB(μV) – 56 dB(μV) quasi-peak 56 dB(μV) – 46 dB(μV) average 0,5 MHz – 5 MHz 56 dB(μV) quasi-peak 46 dB(μV) average 5 MHz – 30 MHz 60 dB(μV) quasi-peak 50 dB(μV) average

3.2.1.3 Test Procedure

The test equipment was placed on a wooden table, and 40cm away from the conductive chamber wall. The EUT was brought into operational mode in a stimulated scenario which favours high emissions (no load, high smps du/dt). Under these operational conditions the AC inlet power port was measured for conducted emissions.

3.2.1.4 Test Verdict

PASSED

3.2.1.5 Used Equipment

Equipment	Make	Model	Calibration
LISN	TEKBOX	TBL5008-2C	2026-MAR
Spectrum Analyzer	SIGLENT	SSA3032X	2024
High pass filter	Tekbox	TBHPF1-150kHz	N/A
HSAR Chamber	INDES EMC	3,5m/2m/1,1m	N/A
EMC Software	Tekbox	EMCVIEW V10.02	N/A

3.2.1.6 Remark

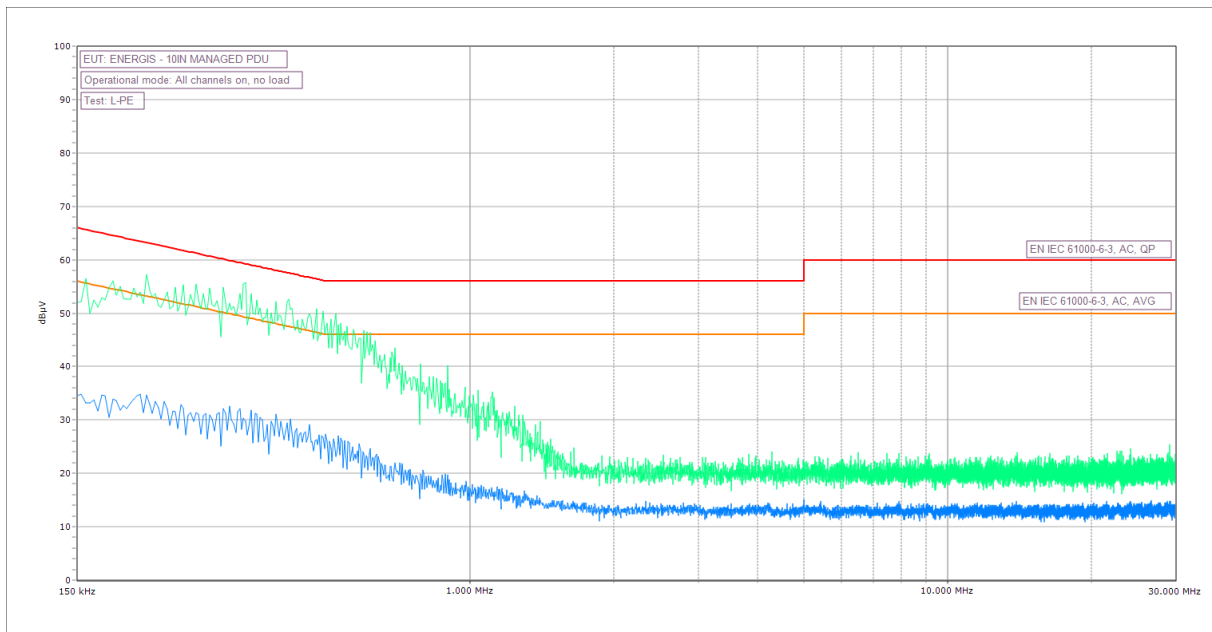


Figure 6: Conducted emissions L-PE

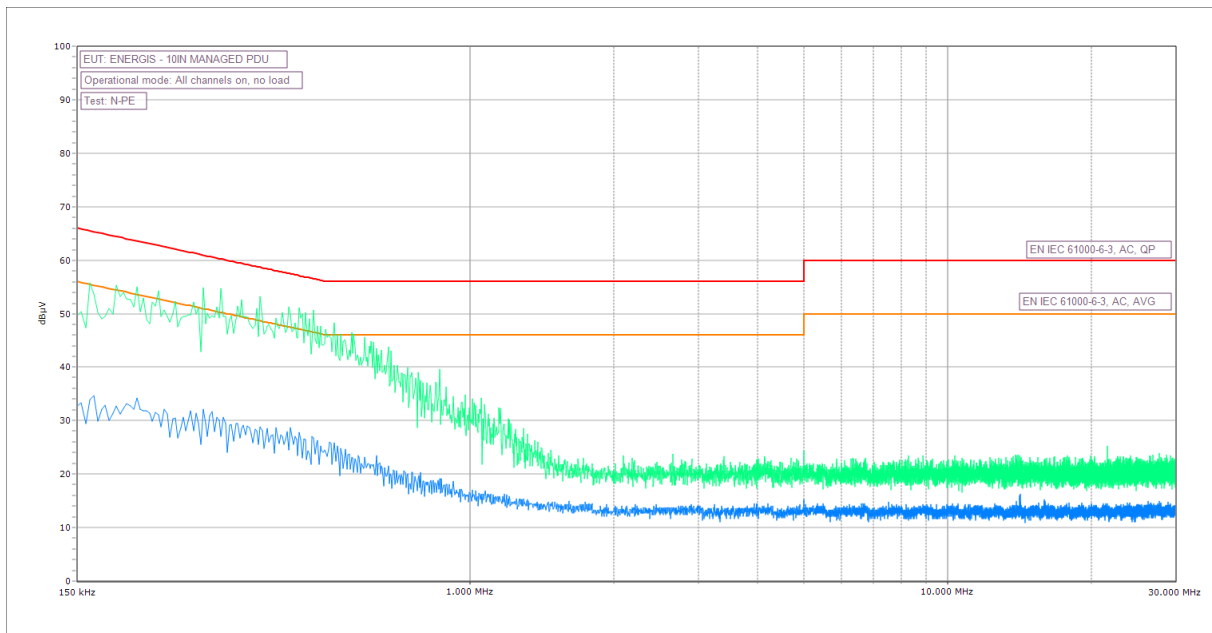


Figure 7: Conducted emissions N-PE

3.2.2 RE1 – Radiated Emissions

3.2.2.1 Test Setup

See 1.2.2 Radiated Emission Setup 30MHz – 1000MHz for test setup arrangement.

3.2.2.2 Test Level

Test Method	Level
Radiated emission over antenna, 3m distance in chamber	30 – 230 MHz: 40dB(μV/m) QP 230 – 1000 MHz: 47dB(μV/m) QP

3.2.2.3 Test Procedure

The test equipment was placed on a wooden table, and 3m away from the measurement antenna. The EUT was brought into operational mode in a stimulus scenario which according favors high emissions.

Under these operational conditions the EUT was measured for radiated emissions.

3.2.2.4 Test Verdict

PASSED

3.2.2.5 Used Equipment

Equipment	Make	Model	Calibration
Spectrum Analyzer	SIGLENT	SSA3032X	2024
High pass filter	Tekbox	TBHPF1-150kHz	N/A
HSAR Chamber	INDES EMC	3,5m/2m/1,1m	N/A
Power supply 24V	Delta Elektronika	SM6020	N/A
Power supply 60V	Delta Elektronika	SM6020	N/A
Biconical Antenna	Tekbox	TBMA1B	2025
EMC Software	Tekbox	EMCVIEW V10.02	N/A

3.2.2.6 Remark

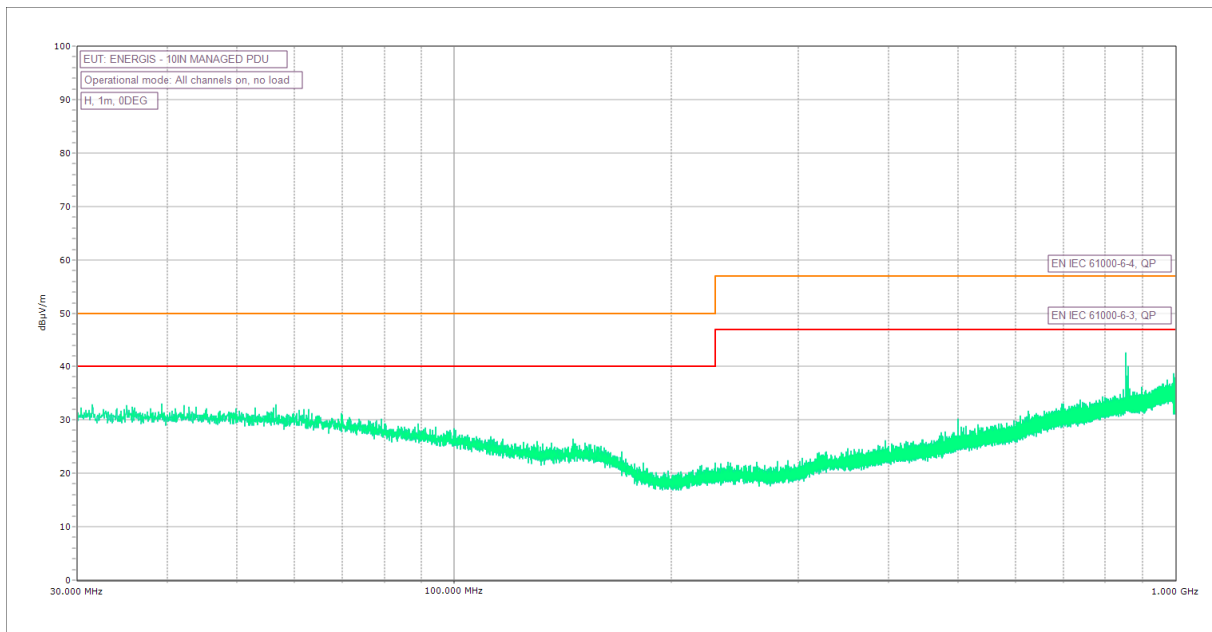


Figure 8: Radiated emission plot - H_1m_0_DEG

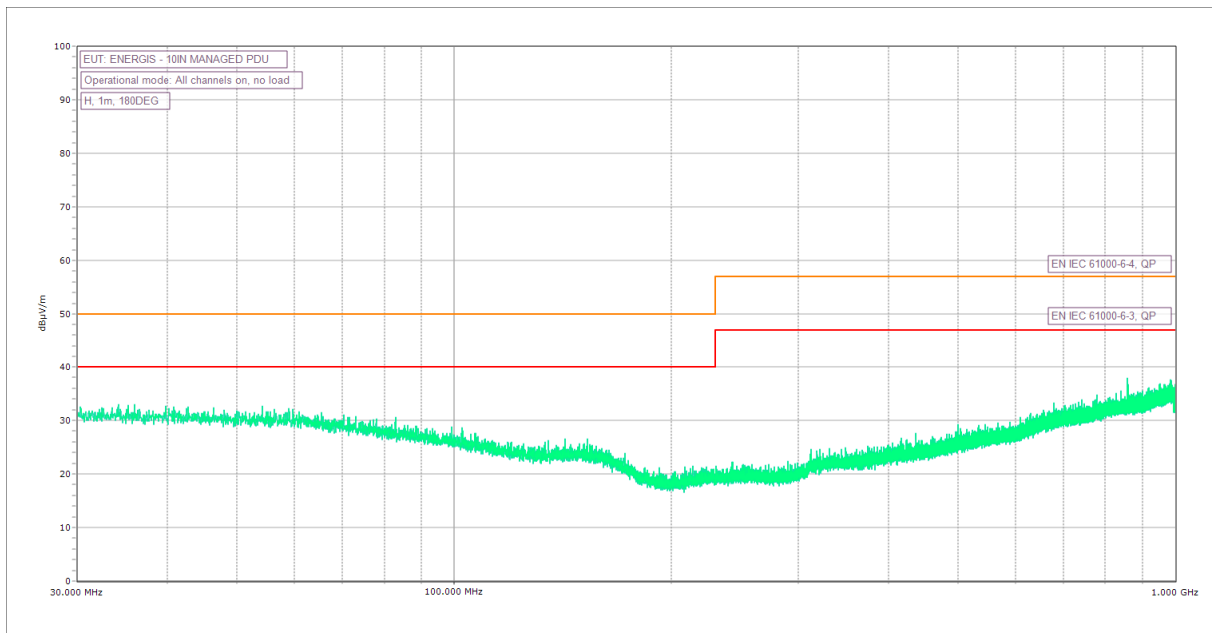


Figure 9: Radiated emission plot - H_1m_180_DEG

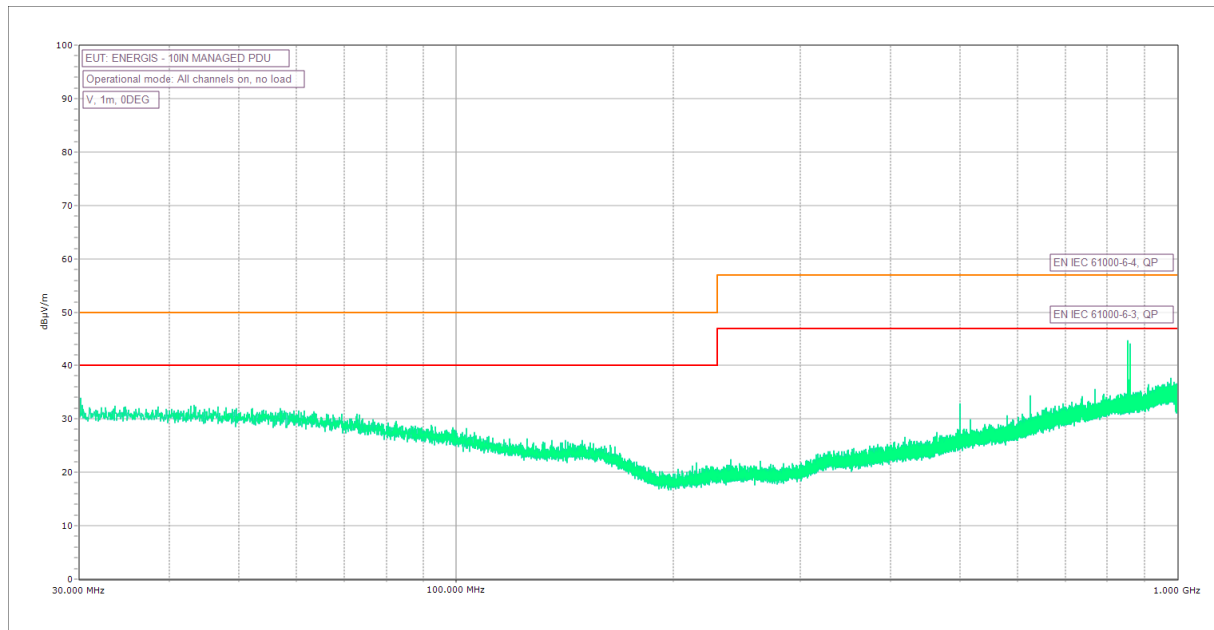


Figure 10: Radiated emission plot - V_1m_0_DEG

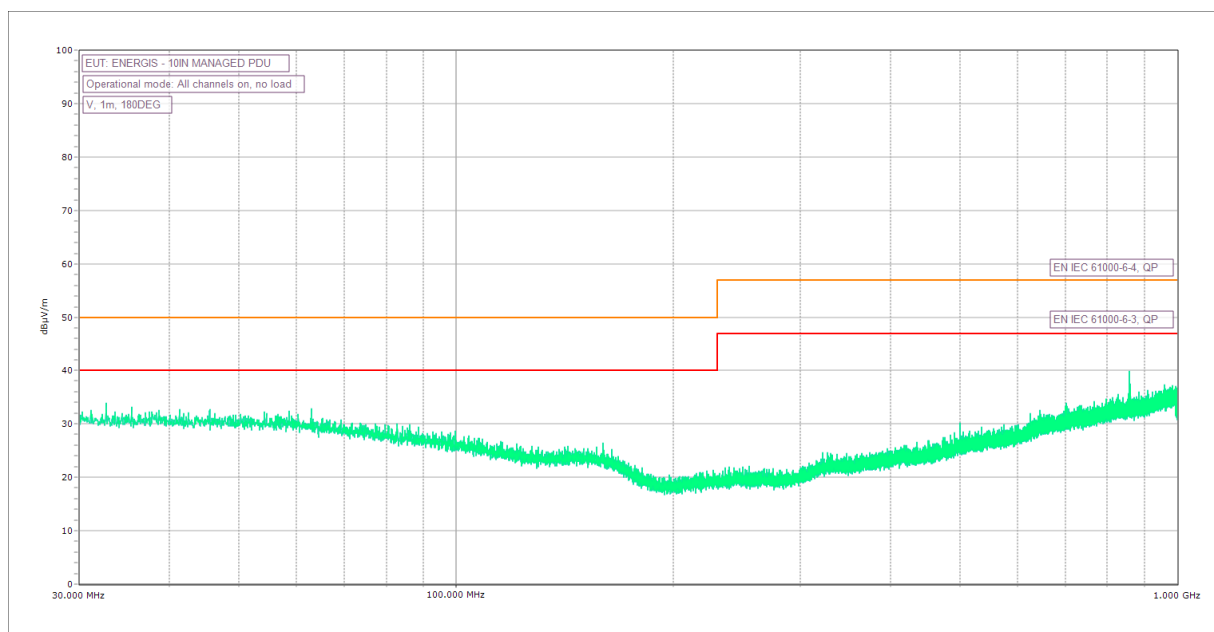


Figure 11: Radiated emission plot - V_1m_180_DEG

4. Test conditions

4.1 Environmental conditions

The tests were carried out under the following climatic conditions:

Temperature: 23° +/-3°

Relative humidity: 27% +/-5%

Atmospheric pressure: 1013.6 hPa

4.2 Auxilliary Test Equipment configuration

Monitoring and Test host device: N/A

4.3 Auxilliary Test Software and stimulus configuration

Test software: N/A

Test software version: N/A

4.4 EUT Firmware version and application code revision

EUT FW identification: N/A

EUT FW revision: N/A

EUT app identification: N/A

EUT app revision: N/A

5. References

[1] EMC Directive 2014/30/EU

Directive of the European Parliament and of the Council on the harmonisation of the laws of the Member States relating to electromagnetic compatibility.

[2] Commission Notice on the Application of the EMC Directive 2014/30/EU

Official guidance document outlining manufacturer obligations, conformity assessment, and technical documentation requirements.

[3] IEC 61000-6-3

Electromagnetic compatibility (EMC) – Part 6-3: Generic standards – Emission standard for residential, commercial and light-industrial environments

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ANNEX I – Foto Documentation





ANNEX II – EMC Chamber statement

EMC Directive 2014/30/EU – Annex II Self-Assessment Justification

Company: INDES EMC e.U.

Address: Zetschegasse 3, 1230 Vienna, Austria

Owner/CEO: Ing. Aleksandar Sindrak

Test Environment Description

Radiated emission measurements were conducted in a shielded enclosure constructed from grounded aluminum panels and conductive fabric layers. The enclosure includes RF absorbers (such as graphene loaded foam), yet without deploying ferrite tiles, resulting in a highly reflective internal environment for frequencies below 200MHz.

Impact on Measurement Conditions

The absence of ferrite absorbers increases internal reflections, which can lead to constructive interference and standing waves. This effect typically elevates the measured radiated emission levels compared to a standard absorber-lined chamber. Therefore, the test environment represents a conservative condition, as emissions are more likely to appear higher than in a compliant chamber.

Rationale for Compliance Assessment

Since the device under test (DUT) meets the applicable radiated emission limits under these worst-case reflective conditions, it is reasonable to conclude that the DUT would also comply in a standard EMC chamber equipped with ferrite tile absorbers, where reflections and resonances are significantly reduced.

Conclusion

Testing in a highly reflective environment without absorbers provides a stringent assessment. Passing under these conditions demonstrates compliance with the EMC Directive requirements and ensures that performance in a fully equipped chamber would not exceed the specified limits.

ANNEX III – Legal Disclaimer for EMC verification support

Disclaimer:

The services provided by *INDES EMC e.U.* in relation to Electromagnetic Compatibility (EMC) are limited to **technical verification, testing assistance, documentation support, and general guidance** intended to help manufacturers fulfill their obligations under the **EMC Directive 2014/30/EU**. These services are **not** a substitute for a manufacturer's own legal responsibilities.

Under **EMC Directive 2014/30/EU**, manufacturers may perform a **self-assessment procedure** to demonstrate conformity, provided they compile technical documentation and ensure their equipment meets the essential protection requirements defined in **Annex I**. The directive allows manufacturers to use **external test laboratories, consultants, or technical experts** to carry out measurements, verifications, or analysis in support of this process. However, the **manufacturer retains full legal responsibility** for ensuring conformity, preparing the technical file, issuing the EU Declaration of Conformity, and affixing the CE marking.

Accordingly, *INDES EMC e.U.* provides support that is strictly **technical and advisory in nature**. All verification activities, measurement data, or technical opinions supplied by *INDES EMC e.U.* are intended solely to assist the manufacturer in performing their own conformity assessment under the applicable modules of the EMC Directive. No service provided constitutes certification, approval, or a legally binding conformity assessment decision. *INDES EMC e.U.* does not assume or accept any responsibility for final compliance of the product, nor for any decisions taken by the manufacturer regarding CE marking, declarations, or market placement.

Manufacturers using these services acknowledge that they remain solely responsible for:

- Ensuring overall conformity of their equipment with **EMC Directive 2014/30/EU**;
- Completing and maintaining the **technical documentation** as specified in Annex II;
- Determining the appropriateness of any test results or verification data; and
- Issuing the **EU Declaration of Conformity**.

By engaging *INDES EMC e.U.*, the manufacturer agrees that any information, recommendations, or support provided are advisory and that ultimate compliance decisions rest exclusively with the manufacturer.

END OF REPORT